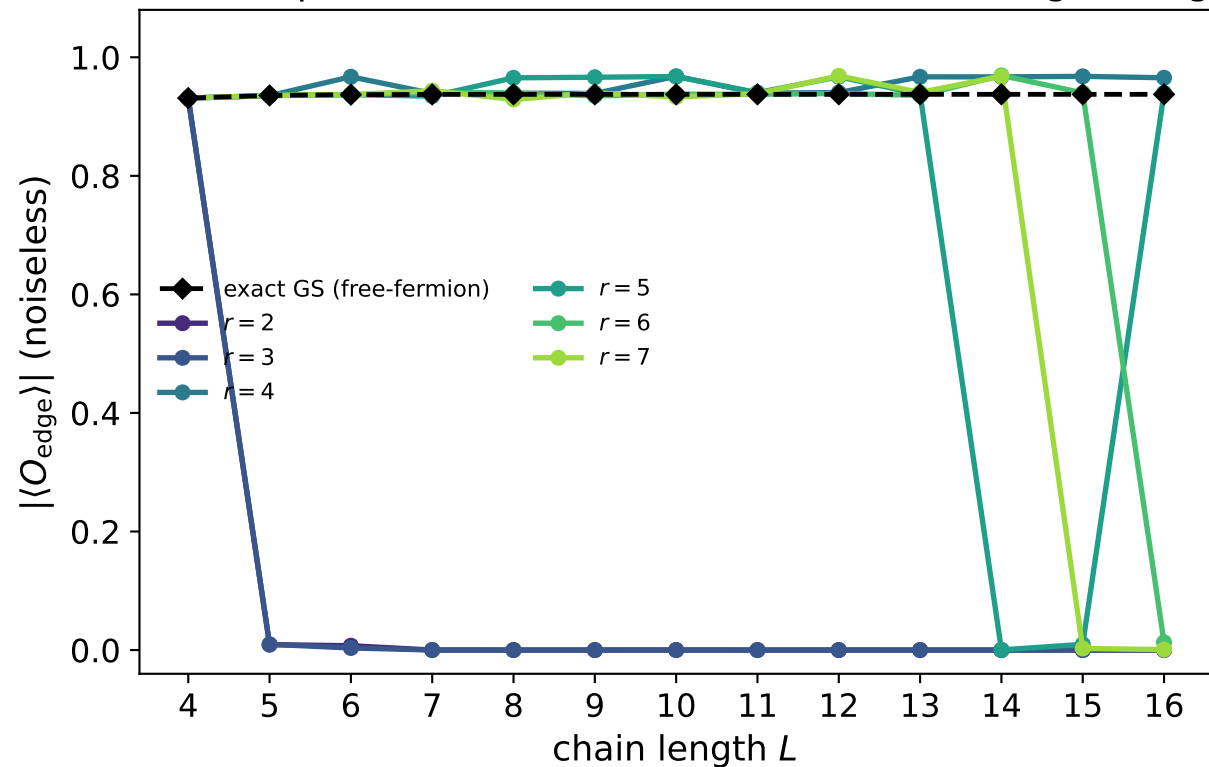


Two ways a fixed-depth preparation fails as L grows ($\mu = 0.5t$, noise: toy depol. $p_{cx} = 0.05$)

Fixed depth: shallow circuits cannot build the edge string



Minimal adequate depth: noise then wins

